TikZ
od zera do elipsy

Mateusz (czarny) Winiarski & ChatGPT

KMPS UJ, WFAIS

dzisiaj

Prezentacja

Link do prezentacji:
`https://github.com/NKFSUJ/LaTeX2025/tree/master/tikz/tikz.pdf`

Co to Ti*k*Z

- Ti*k*Z ist *keine* Zeichenprogramm

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- Ti*k*Z ist *keine* Zeichenprogramm
  - Ti*k*Z to *nie* program do rysowania
- więc Ti*k*Z to biblioteka (mniej więcej) w LaTeXu do robienia obrazków (samemu!)

Skąd się tego nauczyć?

tak jak zawsze:
szukamy w ulubionej wyszukiwarce internetowej `ctan tikz`...

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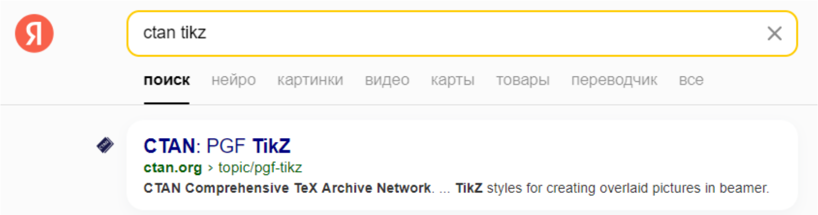

Skąd się tego nauczyć?

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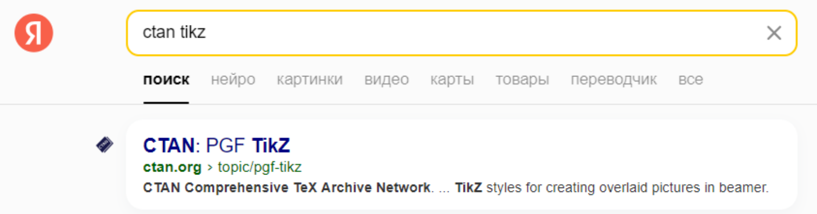
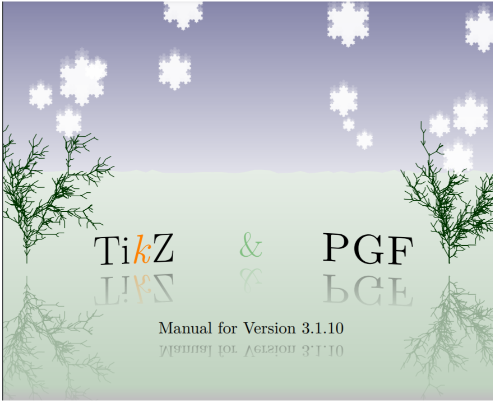


klikamy w pierwszy link...[1]

[1] to nie jest ten link. to nie jest również moja ulubiona wyszukiwarka

...

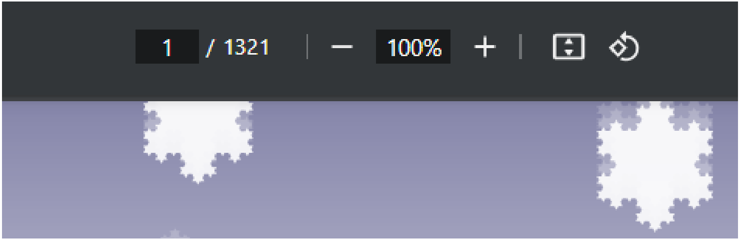
...otwieramy dokumentację...



Rysunek: ładny rysunek

...

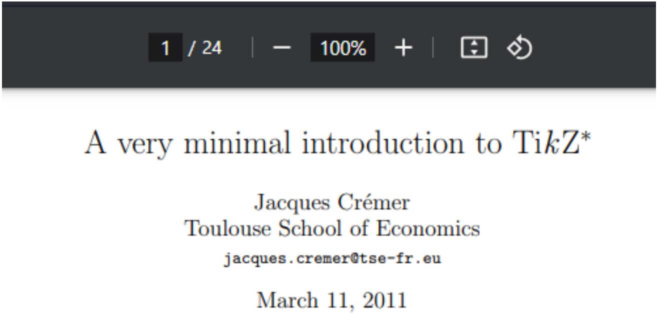
...płaczemy...



Rysunek: 

...

...szukamy lepszego pdfa



A very minimal introduction to TikZ*

Jacques Crémer
Toulouse School of Economics
jacques.cremer@tse-fr.eu

March 11, 2011

Rysunek: Z tego `pdfa` się uczyłem[2], i z niego będę uczył was

[2] ~~tydzień~~ rok temu

Ale ja nie rozumiem...

Internet is your friend
w szczególności:

- ChatGPT
- TEX Stack Exchange

Jak zacząć?

w preambule:

```
\usepackage{tikz}
```

w dokumencie:

```
\begin{tikzpicture}
    % wpisz tu swój rysunek...
\end{tikzpicture}
```

Jak zacząć?

w preambule:

```
\usepackage{tikz}
```

w dokumencie:

```
\begin{tikzpicture}
    % wpisz tu swój rysunek...
\end{tikzpicture}
```

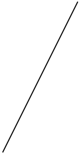
warto rozważyć:

```
\documentclass[margin=5mm]{standalone}
```

Hello, world!

```
\begin{tikzpicture}
    \draw (0,0) -- (1,2);
\end{tikzpicture}
```

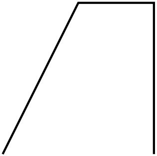
czytaj jako:
TikZie, narysuj mi kreskę od $(0,0)$ *do* $(1,2)$*;*
wynik:



Ale fajne, chcę więcej

```
\begin{tikzpicture}
    \draw[thick] (0,0) -- (1,2) -- (2,2) -- (2,0);
\end{tikzpicture}
```

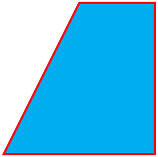
czytaj jako:
TikZie, narysuj mi ***tłustą*** *kreskę od* $(0,0)$ *do* $(1,2)$, *potem do* $(2,2)$, *potem do* $(2,0)$*;*
wynik:



Wincyj!

```
\begin{tikzpicture}
    \draw[thick, red, fill=cyan]
        (0,0) -- (1,2) -- (2,2) -- (2,0) -- cycle;
\end{tikzpicture}
```

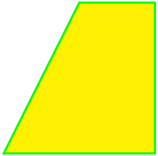
czytaj jako:
TikZie, narysuj mi ***czerwoną tłustą*** *kreskę od* $(0,0)$ *do* $(1,2)$*, potem do* $(2,2)$*, potem do* $(2,0)$*, potem wróć, a wypełnij cyjanowo;*
wynik:



Wincyj!!!

```
\begin{tikzpicture}
    \draw[thick, green, fill=yellow]
        (0,0) -- ++(1,2) -- ++(1,0) -- ++(0,-2) -- cycle;
\end{tikzpicture}
```

czytaj jako:
TikZie, narysuj mi ***zieloną tłustą*** *kreskę od* $(0,0)$*, idź o wektor* $(1,2)$*, potem o wiktor* $(1,0)$*, potem o wektor* $(0,-2)$*, potem wróć, a wypełnij rzułto;*
wynik:



To widzisz już ogólną tendencję:

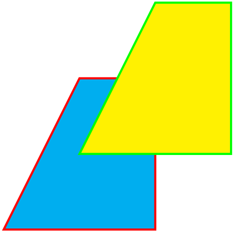
```
\draw[opcje...] (punkt1) figura (punkt2);
```

czytaj jako:
TikZie, narysuj mi figurę od punkt1 do punkt2 z podanymi opcjami;

Czy mogę dać więcej na jednym rysunku?

Tak, i to dokładnie w taki sposób w jaki się domyślasz:

```
\begin{tikzpicture}
    \draw[thick, red, fill=cyan] (0,0) -- (1,2) -- (2,2)
        -- (2,0) -- cycle;
    \draw[thick, green, fill=yellow]
        (1,1) -- ++(1,2) -- ++(1,0) -- ++(0,-2) -- cycle;
\end{tikzpicture}
```



Kolorki

Wiemy już, że kolory możemy definiować tak:
`orange` pomarańczowy
oraz tak:
`\definecolor{neworange}{RGB}{255,140,0}` pomarańczowy

Kolorki

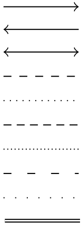
Wiemy już, że kolory możemy definiować tak:
`orange` pomarańczowy
oraz tak:
`\definecolor{neworange}{RGB}{255,140,0}` pomarańczowy
ale jeszcze nie wiemy, że możemy tak:
`orange!80!black` pomarańczowy

Grubość linii

```
\draw[...] (0,0) -- (1,0);
 ultra thin
 very thin
 thin
 semithick
 thick
 very thick
 ultra thick
 line width = 2mm
```

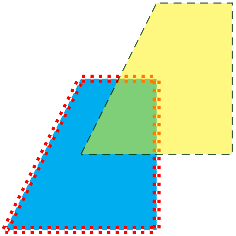
Style linii

```
\draw[...] (0,0) -- (1,0);
 ->
 <-
 <->
 dashed
 dotted
 densely dashed
 densely dotted
 loosely dashed
 loosely dotted
 double
```



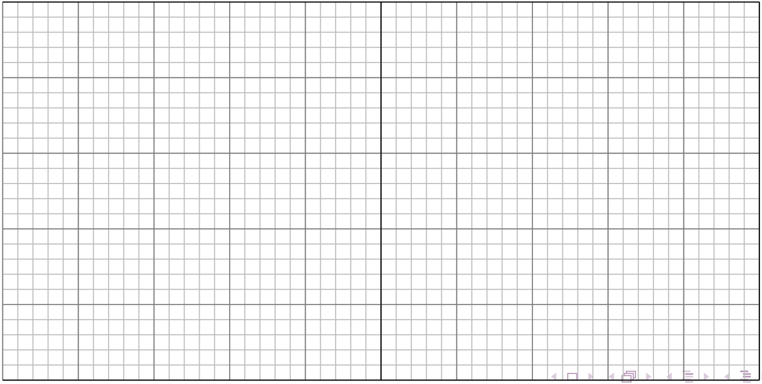
Trochę przykładów

```
\begin{tikzpicture}
    \draw[very thick, red, fill=cyan, dotted, double]
        (0,0) -- (1,2) -- (2,2) -- (2,0) -- cycle;
    \draw[thin, green!20!black, fill=yellow, draw opacity
        = 0.8, fill opacity = 0.5, densely dashed]
        (1,1) -- ++(1,2) -- ++(1,0) -- ++(0,-2) -- cycle;
\end{tikzpicture}
```



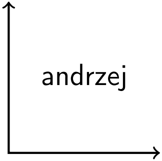
Siatki

```
\begin{tikzpicture}
    \draw[gray!50, step=0.2cm] (-5,0) grid (5,5);
    \draw[gray, step=1cm] (-5,0) grid (5,5);
    \draw[black, step=5cm] (-5,0) grid (5,5);
\end{tikzpicture}
```



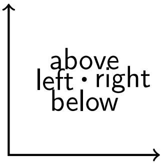
Węzły

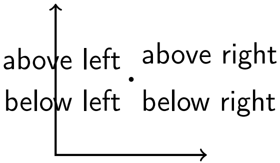
```
\begin{tikzpicture}
    \draw [thick, <->] (0,2) -- (0,0) -- (2,0);
    \node at (1,1) {andrzej};
\end{tikzpicture}
```



Kolanka

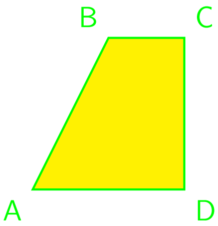
```
\begin{tikzpicture}
    \draw [thick, <->] (0,2) -- (0,0) -- (2,0);
    \draw[fill] (1,1) circle [radius=0.025];
    \node [below] at (1,1) {below};
    \node [above] at (1,1) {above};
    \node [left] at (1,1) {left};
    \node [right] at (1,1) {right};
\end{tikzpicture}
```





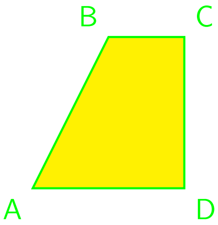
Wincyj kolanek

```
\begin{tikzpicture}
\draw[thick, green, fill=yellow](0,0) node [below left] {A
    } -- ++(1,2) node [above left] {B} -- ++(1,0) node [
    above right] {C} -- ++(0,-2) node [below right] {D} --
     cycle;
\end{tikzpicture}
```



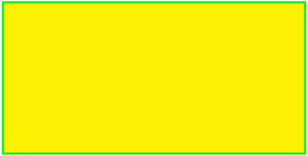
Koordynaty

```
\begin{tikzpicture}
    \coordinate (A) at (0,0);
    \coordinate (B) at (1,2);
    \draw[thick, green, fill=yellow] (A) node [below left]
        {A} -- (B) node [above left] {B} -- ++(1,0) node [
       above right] {C} -- ++(0,-2) node [below right] {D}
        -- cycle;
\end{tikzpicture}
```



Prostokąt

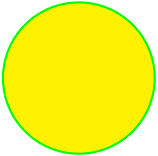
```
\begin{tikzpicture}
\coordinate (A) at (0,0);
\coordinate (B) at (4,2);
    \draw[thick, green, fill=
        yellow] (A) rectangle
          (B) ;
\end{tikzpicture}
```



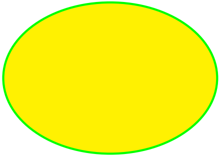
```
\begin{tikzpicture}
\coordinate (A) at (0,0);
\coordinate (B) at (4,2);
    \draw[thick, green, fill=
        yellow, rounded
        corners = 1 cm] (A)
        rectangle (B) ;
\end{tikzpicture}
```

Kółko i elipsa

```
\begin{tikzpicture}
\coordinate (A) at (0,0);
    \draw[thick, green, fill=
        yellow] (A) circle(1
        cm) ;
\end{tikzpicture}
```



```
\begin{tikzpicture}
\coordinate (A) at (0,0);
    \draw[thick, green, fill=
        yellow] (A) ellipse[x
         radius={sqrt(2)}, y
        radius={sin(90)}] ;
\end{tikzpicture}
```



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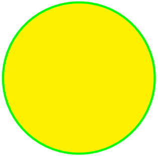
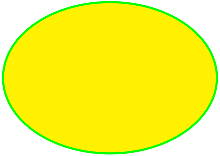
```
\begin{tikzpicture}
\coordinate (A) at (0,0);
    \draw[thick, green, fill=
        yellow] (A) circle(1
        cm) ;
\end{tikzpicture}
```

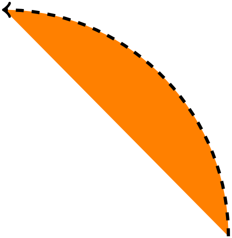


```
\begin{tikzpicture}
\coordinate (A) at (0,0);
    \draw[thick, green, fill=
        yellow] (A) ellipse(
        sqrt(2) and sin(90))
        ;
\end{tikzpicture}
```

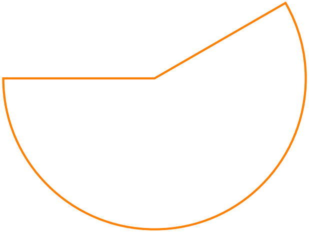


Łuki

```
\begin{tikzpicture}
    \draw[->, dashed, fill=
        orange, very thick]
        (3,0) arc [start
        angle=0, end angle
        =90, radius=3 cm];
\end{tikzpicture}
```

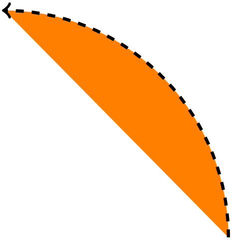


```
\begin{tikzpicture}
    \draw[<->, thick, orange]
        (3,0) arc [start
        angle=180, end angle
        =390, radius=2 cm] --
        (5,0) -- cycle;
\end{tikzpicture}
```

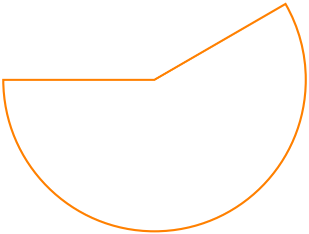


Ł:u:k:i

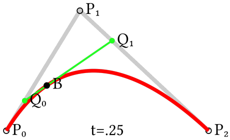
```
\begin{tikzpicture}
    \draw[->, dashed, fill=
        orange, very thick]
        (3,0) arc (0:90:3cm);
\end{tikzpicture}
```

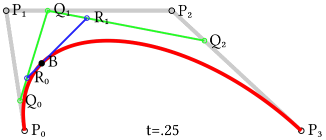


```
\begin{tikzpicture}
    \draw[<->, thick, orange]
        (3,0) arc (180:390:2
      cm) -- (5,0) -- cycle
      ;
\end{tikzpicture}
```



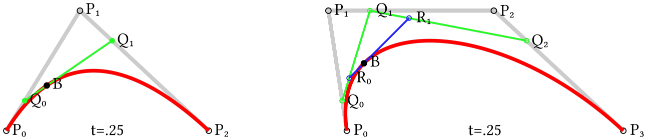
Krzywe Béziera





Rysunek: Rysunek z `https://en.wikipedia.org/wiki/Bezier_curve`

Krzywe Béziera



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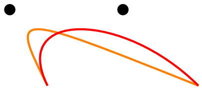
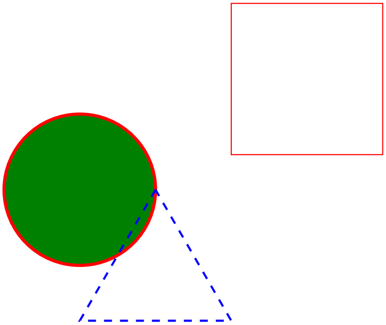
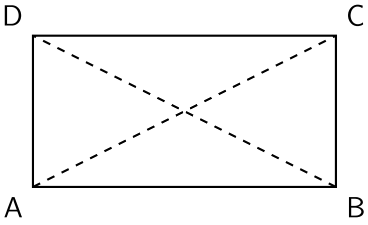


```
\begin{tikzpicture}
    \draw[thick,orange] (0,0) .. controls
        (-0.5,1) .. (2,0);
    \draw[thick,red] (0,0) .. controls
        (-0.5,1) and (1,1) .. (2,0);
    \node at (-0.5,1) {$\bullet$};
    \node at (1,1) {$\bullet$};
\end{tikzpicture}
```

Ćwiczenia I

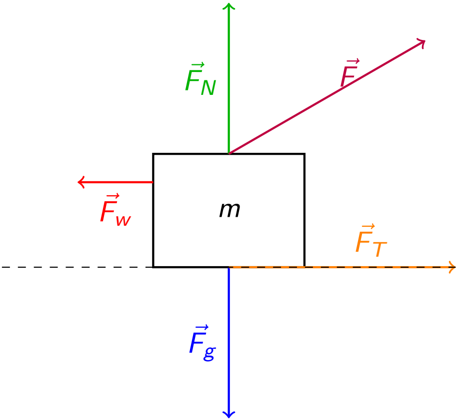


Rysunek: Zadanie 1

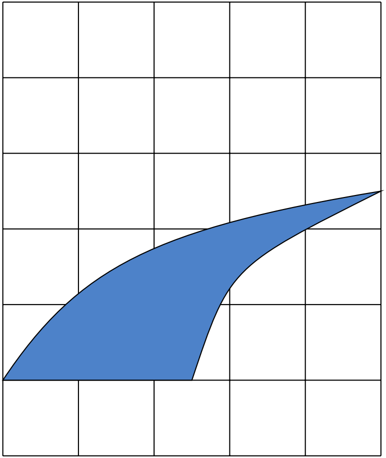


Rysunek: Zadanie 2. Wskazówka: użyj koordynatów

Ćwiczenia II

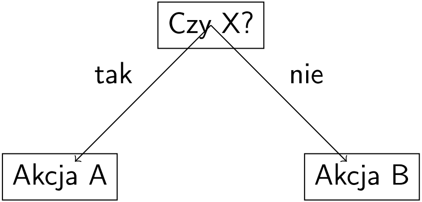


Rysunek: Zadanie 3: magieranika

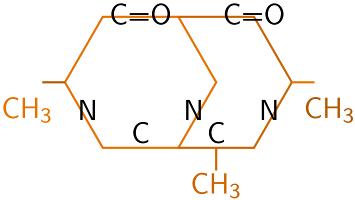


Rysunek: Zadanie 4: przypadkowy kształt

Ćwiczenia III



Rysunek: coś dla informatyków



Rysunek: natalka mnie zabije jak to zobaczy

Rozwiązania

Zadanie 1:

```
\begin{tikzpicture}
    \coordinate (A) at (0, 0);
    \coordinate (B) at (1, 0);
    \coordinate (C) at (2, {4 * sqrt(3)/2 - 3});

    \draw[very thick, red, fill=green!50!black] (A) circle
        (1cm);
    \draw[thick, dashed, blue] (B) -- ++(-1, {-sqrt(3)})
        -- ++(2, 0) -- cycle;
    \draw[thin, red] (C) rectangle ++(2, 2);
\end{tikzpicture}
```

Rozwiązania

Zadanie 2:

```
\begin{tikzpicture}
    \coordinate (A) at (0, 0);
    \coordinate (B) at (4, 0);
    \coordinate (C) at (4, 2);
    \coordinate (D) at (0, 2);

    \draw[thick] (A) -- (B) -- (C) -- (D) -- cycle;

    \draw[thick, dashed] (A) -- (C);
    \draw[thick, dashed] (B) -- (D);

    \node[below left] at (A) {A};
    \node[below right] at (B) {B};
    \node[above right] at (C) {C};
    \node[above left] at (D) {D};
\end{tikzpicture}
```

Rozwiązania

Zadanie 3:

```
\begin{ tikzpicture }
    \def\F{3}       % Dlugosc wektora sily  F
    \def\angle{30} % Kat sily  F (w stopniach)
    \def\m{1.5}    % Wysokosc prostokata

    \draw[thick] (0,0) rectangle (2,\m) node[midway] {$m$};

    \draw[->, thick, blue] (1, 0) -- ++(0, -2) node[midway, left] {$\vec{F}
         _g$};
    \draw[->, thick, green!70!black] (1, \m) -- ++(0, 2) node[midway, left] {$\
         vec{F}_N$};
    \draw[->, thick, red] (0, 0.75*\m) -- ++(-1, 0) node[midway, below] {$\
         vec{F}_w$};
    \draw[->, thick, purple] (1, \m) -- ++({\F*cos(\angle)},{\F*sin(\angle)})
        node[midway, above right] {$\vec{F}$};
    \draw[->, thick, orange] (1, 0) -- ++({\F*cos(0)},{\F*sin(0)})
        node[midway, above right] {$\vec{F}_T$};

    \draw[dashed] (-2, 0) -- (4, 0);
```

Rozwiązania

Zadanie 4:

```
\begin{tikzpicture}
\draw[step=1cm] (0,-1) grid (5,5);
\draw[fill=blue!70!green!70!white]
    (0,0) .. controls (1,1.5) and (2,2) .. (5,2.5)
     .. controls (3,1.5) .. (2.5,0)
     -- cycle
    ;
\end{tikzpicture}
```

Rozwiązania

Zadanie 5:

```
\begin{tikzpicture}
\node at (0,0) [draw, rectangle, shape aspect=2] {Czy X?};
\node at (-2,-2) [draw, rectangle] {Akcja A};
\node at (2,-2) [draw, rectangle] {Akcja B};
\draw[->] (0,0) -- (-1.8,-1.8) node[midway, above left] {
    tak};
\draw[->] (0,0) -- (1.8,-1.8) node[midway, above right] {
    nie};
\end{tikzpicture}
```

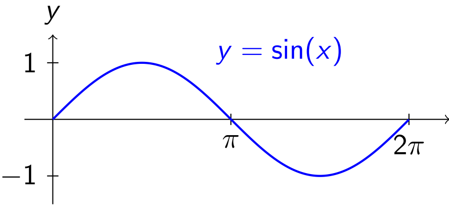
Rozwiązania

Zadanie 6:

```
\begin{ tikzpicture }
\def\cola{orange!90!black}
\def\colb{orange!80!black}
\def\colc{orange!70!black}

\draw[thick, \cola] (0,0) -- (1,0) -- (1.5,0.866) -- (1,1.732) -- (0,1.732)
    -- (-0.5,0.866) -- cycle;
\draw[thick, \colb] (1,0) -- (2,0) -- (2.5,0.866) -- (2,1.732) -- (1,1.732);

\node at (-0.2,0.5) {N};
\node at (1.2,0.5) {N};
\node at (2.2,0.5) {N};

\node at (0.5,0.2) {C};
\node at (1.5,0.2) {C};
\node[above] at (0.5,1.5) {C=O};
\node[above] at (2,1.5) {C=O};

\node[text=\cola] at (-1,0.5) {CH$_3$};
\node[text=\colb] at (1.5,-0.5) {CH$_3$};
\node[text=\colc] at (3,0.5) {CH$_3$};

\draw[thick, \colc] (-0.5,0.866) -- (-0.8,0.866);
\draw[thick, \colb] (1.5,0) -- (1.5,-0.3);
```

Wykres I

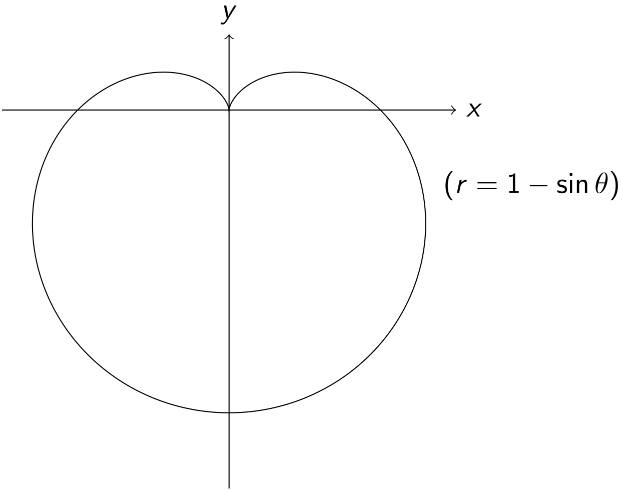
```
\begin{ tikzpicture }
    % Osie  ukadu      wsprzdnych
    \draw[->] (-0.5,0) -- (7,0) node[
         right] {$x$};
    \draw[->] (0,-1.5) -- (0,1.5) node[
         above] {$y$};
    % Etykiety  osi
    \draw  (3.14,0.1)  -- (3.14,-0.1)
         node[below] {$\pi$};
    \draw  (6.28,0.1)  -- (6.28,-0.1)
         node[below] {$2\pi$};
    \draw  (0.1,1)  -- (-0.1,1) node[left]
          {$1$};
    \draw (0.1,-1) -- (-0.1,-1) node[
         left] {$-1$};
    % Wykres funkcji  sin (x)
    \draw[thick ,  blue ,  samples=100,
         domain=0:6.28, smooth]
        plot  (\x,{ sin (\x  r )});
    % Opis wykresu
    \node[blue]  at  (4,1.2)  {$y = \sin(x)
         $};
\end{ tikzpicture }
```



Wykres II

```
\begin{figure}
\centering
\begin{tikzpicture}[scale=2]
    \draw[->] (-1.5,0) -- (1.5,0) node[right] {$x$};
    \draw[->] (0,-2.5) -- (0,0.5) node[above] {$y$};
    \draw[domain=0:360,samples=200,smooth,variable=\t]
        plot ({(1 - sin(\t)) * cos(\t)}, {(1 - sin(\t)) *
            sin(\t)});
    \node at (2,-0.5) {$(r = 1 - \sin\theta)$};
\end{tikzpicture}
\caption{Krzywa Huwdu w \Tikz{}ie}
\end{figure}
```

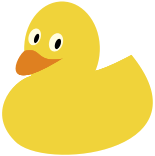
Wykres II



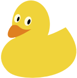
Rysunek: Krzywa Huwdu w TikZie

Kaczki!

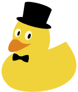
```
% \usepackage{tikzducks}
\duck
```



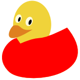
Kaczki!!



`\duck`



`\duck[tophat,bowtie]`



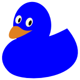
`\duck[tshirt=red]`

`\duck[longhair=blond]`

`\duck[glasses]`

`\duck[umbrella=blue!50!b`

`\duck[cap=red]`



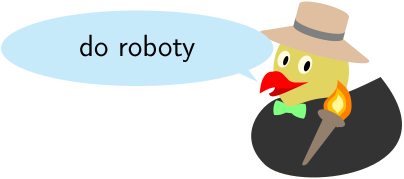
`\duck[body=blue]`

`\duck[santa,beard]`

`\randuck`

ctan tikzducks

https://sunsite.icm.edu.pl/pub/CTAN/graphics/pgf/contrib/tikzducks/tik



https://overleaf.uj.edu.pl/read/wsbzfrgtcvnc